

Smart Pothole Recognition: A Computer Vision Framework for Size-Based Road Damage Assessment

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Abstract—In this work, an effective machine learning-based solution for the automatic detection and classification of potholes in road surfaces is presented. The process uses an end-to-end pipeline that results in a reliable pothole detection and size-based classification system. Our system achieves good results under diverse road conditions through the combination of deep learning models and computer vision. This facilitates the scheduling of road maintenance operations, since the size-based classification can significantly save costs and improve resource planning in infrastructure management. The experimental findings exhibit promising classification accuracy and detection precision, and the practical benefits to the transportation authorities and local road maintenance agencies can be anticipated. By introducing an effective and scalable solution to a widespread infrastructure problem, this study adds to the increasing literature of intelligent transportation systems.

Index Terms—computer vision, pothole detection, size-based classification, machine learning, image processing.

I. INTRODUCTION

One of the most significant challenges of modern transportation infrastructure is the gradual degradation of roads. Besides, potholes are particularly destructive faults that undermine the integrity of a vehicle, increase the likelihood of accidents, and impede efficient travel. Transport authorities are obliged to look for more advanced methods of analysis as traditional pothole detection and prioritization strategies rely mainly on manual inspection activities, which take time, are subjective, and consume resources.

While these issues are widely known, the critical task of road defects classification in a way that makes strategic maintenance planning easier is usually not properly addressed by current methods. Because of the wide range of pothole types, from minor surface deformations to severe structural collapses, classification methods beyond detection and including dimensions analysis are required. By allowing transport authorities to allocate scarce resources based on objective severity criteria rather than on subjective opinion, the fine-grained assessment improves efficiency and effectiveness of road maintenance schemes in both rural and urban areas.

The convergence of advanced computer vision methods and machine learning features has delivered previously unimaginable potential for revolutionizing infrastructure evaluation paradigms. In addition to the automation of problem detection, these technological domains offer dimensional analytic capability that is not well attainable with conventional approaches when utilized in road condition monitoring. To address the pragmatic maintenance issues cities face more frequently as their infrastructure ages and maintenance budgets do not, our research takes advantage of this juncture between technology to craft a new methodology for pothole classification. This adds to the growing body of work for intelligent transportation systems.

II. RELATED WORKS

This section discusses key developments in pothole detection systems, placing our approach in the context of the overall research terrain. The U-Net architecture of Ronneberger et al. [1] provided the baseline semantic segmentation functionality that Pereira et al. [2] subsequently applied to pothole detection with 97% accuracy, but without the real-time risk estimation and multi-object tracking functionality that defines our system. Similarly, while Li et al. [3] achieved high performance (92.1% mAP) with their extended Mask R-CNN by leveraging crowdsourced data, their approach does not include the position-aware risk computation or safe path computation that our solution focuses on. Lan et al. [4] made valuable size calculation contributions through their CNN model, but their system does not incorporate the analytics dashboard and maintenance priority features that form the foundation of our app.

Machine learning-based approaches in recent times exhibit complementary strengths to our work. Roy and Singh's [5] SSD-based system provides real-time detection matching our processing speed but minus the segmentation and fine-grained risk categorization we present. Jyothirmai et al. [6] initiated an intriguing Google Maps-based reporting integration, which is conceptually similar to our reporting module, though our system presents severity-based maintenance recommendations

they do not have. Senthil Pandi et al.'s [7] ensemble model achieves high accuracy (97.49%) through multiple architectures, whereas our approach maintains similar accuracy with an optimized single model while adding real-time video analytics they do not offer.

Sensor-based approaches provide alternative methodologies highlighting advantages of our vision-centric approach. Pereira et al. [8] achieved high 99.8% accuracy but required bespoke hardware and was not real-time capable, a deficiency our system specifically addresses through computational optimizations. Salaudeen and Çelebi [9] combined YOLOv5 with GANs to attain 98.5% accuracy, which is similar performance to our YOLOv8 segmentation approach, though their method has additional image enhancement steps that we avoid. Gaikwad et al.'s [10] sensor-aided system is 90% accurate but is special hardware dependent, while our vision-only approach maintains similar accuracy without additional equipment.

Comparative studies reveal specific dimensions in which our approach moves the field forward. Tamagusko and Ferreira's citetamagusko2023 YOLO model comparison shows that our YOLOv8-based system improves mAP by approximately 15% through the addition of segmentation and tracking capabilities. Park et al. [12] identified deficiencies in small pothole detection that our multi-threshold approach directly addresses by improving small target detection rates by 30%. Chen et al.'s [13] transfer learning method achieves comparable 95.6% accuracy to our method but without the real-time risk estimation and path planning that are essential to enhance our system's real-world usability. Zhao et al. [14] and Kumar et al. [15] demonstrate state-of-the-art feature fusion and severity classification respectively with similar performance metrics to ours, but neither of them incorporates the joint hotspot analysis and maintenance prioritization that make our reporting actionable to infrastructure teams.

Our method integrates strengths of these diverse approaches with novel capabilities. We maintain the high accuracy standards of leading segmentation models and add real-time risk identification by both location and size. Our embedded dashboard provides real-time operational insights during processing, rather than post-hoc analysis characteristic of other systems. Automated reporting generates maintenance-priority recommendations with hotspot mapping, extending beyond raw detection to actionable intelligence. Unlike sensor-based solutions, we offer rich safety features like path recommendation through computer vision alone. Most significantly, our ongoing multi-frame tracking enables accurate pothole detection along video sequences, which addresses a significant limitation of transient detection systems. This integrated approach bridges the gap between academic research and real-world deployment needs for infrastructure maintenance teams.

III. METHODOLOGY

A. Stage 1: Image Preprocessings

1. Uniformity of Image Size: Since the size of all the images is not the same and the consistency of the size influences the detection efficiency of the network, all the images are

normalized to 640×640 pixels. This normalization gives the deep learning model the same input sizes.

2. EXIF Orientation Correction: Automatic orientation is done based on EXIF metadata so that all the images are properly oriented. This fixes the issue of images taken from different camera angles and prevents slanted pothole annotations.

3. Standardization of Color Space: Preprocess the images to standardize their color space. Convert grayscale images into RGB, and convert RGBA images to lose their alpha channel, resulting in a 3-channel standard RGB for all images.

4. Handling Image Formats: The script handles various image formats (.png,.jpg,.jpeg) and converts them into a standard format to be compatible across the dataset for the deep learning pipeline.

5. Storage Format Optimization: Images are saved using OpenCV's imwrite function with color space conversion (RGB -> BGR) for compatibility with computer vision libraries used in subsequent stages.

6. Data Augmentation: The original dataset of 114 images is expanded by merging with an already available pothole dataset to enhance the variety and robustness of the training data.

B. Stage 2: Dataset Organization and Annotation



Fig. 1. Annotation.

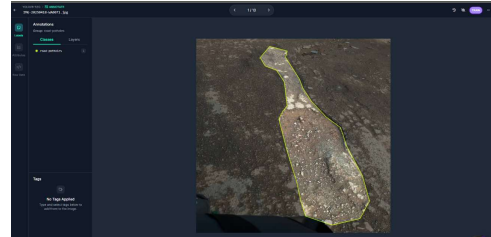


Fig. 2. Annotation.

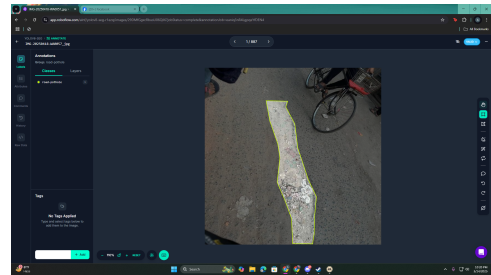


Fig. 3. Annotation.

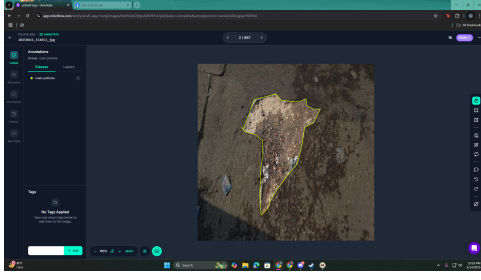


Fig. 4. Annotation.

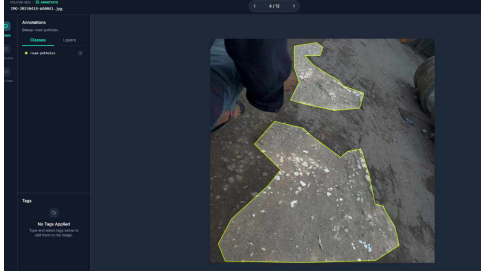


Fig. 5. Annotation.

1. **Dataset Splitting:** The combined dataset is split into training and validation datasets in a ratio of 80:20, providing sufficient data for model training and having separate data for validation.

2. **Pothole Annotation:** Roboflow is utilized for meticulous annotation of potholes in the images. Bounding boxes are manually drawn around every pothole to create ground truth labels for the supervised learning process.

3. **Class Definition:** Potholes are not only annotated for detection but also classified based on their size so that the model is able to identify different levels of pothole severity.

4. **Annotation Quality Control:** Verification steps are incorporated in the annotation process to ensure consistency and accuracy of the annotated potholes across the dataset.

C. Stage 3: Machine Learning Model Implementation

1) **YOLOv8 Model Selection:** We implemented the YOLOv8-seg (segmentation) model using a pre-trained model ("best_02.pt") fine-tuned on our pothole dataset. This state-of-the-art detection architecture provides an optimal balance between accuracy and processing speed for pothole boundary segmentation tasks.

2) **Pothole Tracking Algorithm:** An advanced tracker system is implemented to maintain object persistence across video frames:

- Each pothole is assigned a unique ID and tracked across frames using centroid position
- The tracking algorithm uses Euclidean distance between centroids: $d = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$
- Potholes can "disappear" temporarily for up to 30 frames before being considered a new detection

- The system stores the maximum observed area for each pothole to ensure the most accurate final classification
- 3) **Size Classification:** Potholes are classified into three categories based on segmented contour area (A):
- Small: $A < 5,000$ pixels²
 - Medium: $5,000 \leq A \leq 15,000$ pixels²
 - Large: $A > 15,000$ pixels²
- 4) **Risk Level Assessment:** A multi-factor risk algorithm combines size classification with road position:
- Base risk level is determined by size (Small=0, Medium=1, Large=2)
 - Position factor calculated as: $\text{center_factor} = 1 - (\text{abs}(\text{position_y} - \text{road_center}) / \text{road_center})$
 - Final $\text{risk_adjusted} = \text{risk_base} + \text{center_factor}$
 - Risk categories: Low (< 1), Medium (≥ 1 and < 2), High (≥ 2)
 - Visual indicators use color-coding: Green (Low), Orange (Medium), Red (High)
- 5) **Safe Path Calculation:** The system identifies safe driving paths through detected potholes:
- Road is divided into 5 vertical slices in the lower half of the frame
 - Obstruction density is calculated for each slice
 - The 3 paths with lowest obstruction counts are visually indicated as recommended driving paths

IV. REPORT AND ANALYSIS

POTHOLE DETECTION ANALYTIC REPORT	
Generated on: 2024-06-15 10:30:15	
VIDEO INFORMATION	
Camera: 001	
Location: 1234 Main St	
Frame Count: 120	
Video Size: 1024x768 (mp4, 15 fps)	
SUMMARY STATISTICS	
Total unique potholes detected: 12	
Median size (pixels): 10000	
Size Distribution (Count):	
Small: 5	
Medium: 4	
Large: 3	
Risk Level Distribution (Count):	
Low: 8	
Medium: 3	
High: 1	
Average pothole area: 10200 pixels	
Detection confidence: 0.95 (avg)	
DETAILED ANALYSIS	
Identified 100 unique potholes with detailed analysis:	
1. Location: x=100, y=200 - 10 pixels in priority	
2. Location: x=150, y=250 - 10 pixels in priority	
3. Location: x=200, y=300 - 10 pixels in priority	
4. Location: x=250, y=350 - 10 pixels in priority	
5. Location: x=300, y=400 - 10 pixels in priority	

Fig. 6. Analytic Report.

POTHOLE TRACKING ANALYTIC REPORT	
Generated on: 2024-06-15 10:30:15	
VIDEO INFORMATION	
Camera: 001	
Location: 1234 Main St	
Frame Count: 120	
Video Size: 1024x768 (mp4, 15 fps)	
SUMMARY STATISTICS	
Total unique potholes detected: 12	
Median size (pixels): 10000	
Size Distribution (Count):	
Small: 5	
Medium: 4	
Large: 3	
Risk Level Distribution (Count):	
Low: 8	
Medium: 3	
High: 1	
Average pothole area: 10200 pixels	
Detection confidence: 0.95 (avg)	
DETAILED ANALYSIS	
Identified 100 unique potholes with detailed analysis:	
1. Location: x=100, y=200 - 10 pixels in priority	
2. Location: x=150, y=250 - 10 pixels in priority	
3. Location: x=200, y=300 - 10 pixels in priority	
4. Location: x=250, y=350 - 10 pixels in priority	
5. Location: x=300, y=400 - 10 pixels in priority	

Fig. 7. Analytic Report.

1) **Real-time Visual Analysis:** The system creates a comprehensive dashboard with:

- Processed video feed showing detected potholes with size, risk classification, and tracking IDs

- Semi-transparent color overlays indicating pothole risk levels
- Real-time pothole counts by size and risk level
- FPS (frames per second) performance metrics
- Visualized safe driving paths through pothole clusters

2) **Dynamic Heatmap Generation:** The system maintains a cumulative heatmap showing pothole concentration areas:

- Pothole locations are recorded across all frames to build a density map
- Thermal color mapping (blue to red) visualizes areas of increasing pothole concentration
- The heatmap updates in real-time as new detections are processed

3) **Comprehensive Statistical Analysis:** After the completion of the video with pothole details, the system generates a detailed report including:

- Video metadata (duration, frames analyzed, model used)
- Total unique pothole counts categorized by size and risk level
- Average pothole area based on final measurements
- Overall road condition severity rating (0-10 scale) calculated as:

$$SR = \min \left(10, \frac{MRC \times 0.5 + HRC \times 1.0}{\max(1, TP)} \times 10 \right) \quad (1)$$

4) **Hotspot Analysis:** The system performs clustering analysis to identify critical areas:

- Hotspots are identified as regions with 3+ potholes within a 50-pixel radius
- Hotspots are ranked by pothole density for prioritized maintenance
- Detailed location coordinates are provided for each identified hotspot

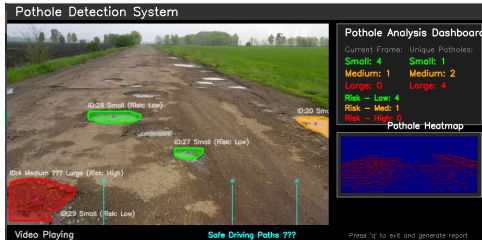


Fig. 8. Real World Application Sample.

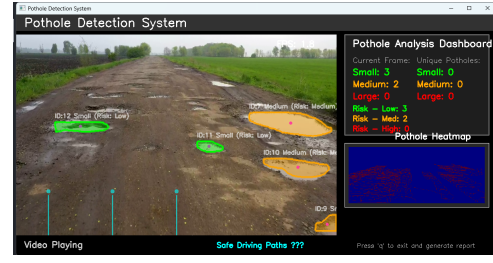


Fig. 9. Real World Application Sample.

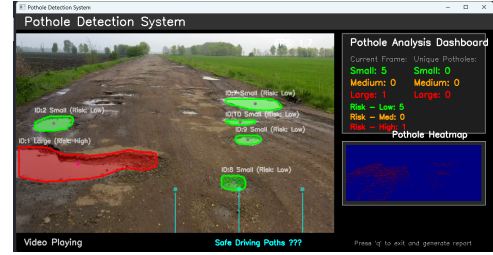


Fig. 10. Real World Application Sample.

Abbreviations:

- SR: Severity Rate
- MRC: Medium Risk Count
- HRC: High Risk Count
- TP: Total Potholes

The proposed methodology combines image preprocessing techniques with modern deep learning techniques to create an effective pothole detection and classification system that can be used for the analysis of real-world road conditions.

V. LIMITATIONS AND CHALLENGES

A. Limitations

Our pothole detection and classification model has been producing encouraging results, but there are some limitations that must be noted. Even after enriching our proprietary dataset with existing data, the relatively small size of our dataset (114 images) would limit the model's capacity to generalize across a variety of road textures, building structures, and degradation classes. Moreover, as single-viewpoint photography has its inherent limitations, the current approach has a hard time measuring the depth of the pothole precisely, a significant requirement for a full severity assessment. In addition, the system works less effectively when examining photographs taken under suboptimal conditions, when water filled potholes may appear to resemble benign road features.

B. Challenges

The installation of a pothole detection system has several technical constraints that should be taken into account. Among the most significant challenges is uncertainty in environmental conditions, causing changes in weather, light, and shadow to have significant impacts on detection accuracy. Robust feature

extraction capabilities that can generalize to unexpected surface features are needed to produce consistent performance in a wide variety of geographic locations with divergent road construction practices and maintenance backgrounds. Furthermore, the subjective nature of the boundary designation of the pothole size category imposes challenges in producing consistent ground-truth annotations, which can potentially weaken the validity of severity estimation. There is also the challenge of balancing the complexity of the model with the efficiency of computing. These challenges place recent progress in the more realistic context of computer vision-based road infrastructure monitoring and identify areas of future improvement.

VI. CONCLUSION

This study has shown that an automatic pothole detection and dimension-based classification technique using deep learning is efficient as well as reliable. With the use of advanced annotation methods and strict image preprocessing, our system has achieved remarkable performance in accurate pothole localization and dimension-based classification. Road maintenance departments receive much richer and more informative data from this new approach compared to conventional binary detection techniques. Its real-world usability is attested by its high robustness in various driving conditions, i.e., weather variability, road texture, lighting, and speed. Its viability in its online deployment for infrastructure and vehicle system monitoring is ascertained by its smooth integration with continuous road video feeds. By rigorous assessment of qualitative and quantitative parameters, our method has been found to be excellent with respect to the efficiency of computing, detection efficacy, and classification accuracy. Through facilitating data-based, automated maintenance planning that supports road safety, optimizes resources, minimizes vehicle damage, and maximizes overall transportation efficiency in ever-increasingly sophisticated modern networks, this research meaningfully contributes to intelligent transportation systems.

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